Design Motivations, Tokenomics, and Scalability

Space Documentation Twin
Challenge 2 Documentation scarcity | Consortium PoA
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1. Design Motivation

1.1 Hardware setup and consensus mechanism

We choose a consortium Proof of Authority network. In space supply chains, trust and compliance matter. Participants are known organisations. A small validator set gives low latency and predictable performance. For a pilot, this is more realistic than a fully open public chain.

Hardware view: user-facing apps connect to RPC gateways. Validators run EVM execution and block production. An indexer listens to events and builds a search layer. Storage nodes pin files on IPFS.

1.2 On-chain vs off-chain data

On-chain We store only the minimal truth: ownership, small hashes, CIDs, prices, payments, and access rights. Off-chain we store large files: JSON metadata and PDF test reports. This is mainly for cost reasons. Also, documents change format and can be extended, so an off-chain JSON is more flexible.

For integrity, the CID binds the content. If someone changes a PDF, the CID changes. The chain still points to the old CID, so the change is detected. For availability, the consortium should run a pinning policy. For very long lifetimes, an Arweave backup is an option.

1.3 The output

The main output is a simple Documentation Completeness Score. It is not a physics model. It is just a quick decision tool. The procurement engineer can compare two parts and see which one has stronger evidence.

2. Tokenomics (users, incentives, sanity)

2.1 Users

We focus on three user types:

- Test house manager: owns radiation test infrastructure. Wants to monetise reports more than once.
- Procurement engineer in a startup: wants faster access to credible evidence, cannot repeat expensive tests.

- QA / reliability lead in a prime contractor: cares about audit trail and risk reduction; can pay for trusted data.

2.2 Utility of the NFT?

The NFT is a digital passport for documentation. It does not replace the physical part. It links a batch identifier to verifiable pointers (CIDs) and a payment history. Compared to the current way such as private emails, NDAs, scattered PDFs etc., this creates a cleaner audit information trail and faster access. However, we cannot ignore the issue of multiple parties' authority, such as how repair shops and testing agencies are authorized and how to obtain responsible signatures.

2.3 Incentives and behaviours to motivate

The behavior we want is simple: more test documentation should be published and discoverable. In a decentralised setting, you cannot force actors, so incentives matter.

Mechanisms used:

- Pay-per-access fee: buyers pay, test houses earn. This makes publication a revenue stream.
- Stake when publishing: the test house locks a stake, so low-quality spam is costly.
- Slashing by arbiter: if fraud is proven, the stake can be reduced. This adds basic accountability.
- Reputation of off-chain indexer: repeat publishers can be ranked in the UI using event history.

Economic sanity check: the buyer should not pay more than the cost of repeating a test. In the pilot, fees can be small. The goal is to prove behaviour change, not to maximise profit.

2.4 Bootstrapping

Early on, the network has few reports, so buyers may not come. This is the bootstrapping problem. For the pilot, the consortium can subsidise early report publication (grants, or fee matching). Once the first automated payments happen, the incentive is visible.

2.5 Optional oracle note

If fees need to stay stable in EUR terms, the system can price access in a stablecoin or use an oracle for conversion. If an oracle affects payments, it must be correct and available, and the system must know who provided the data (accountability).

3. Scalability

3.1 What an initial pilot looks like

Pilot should be small and safe. Example setup: 3 validators (one agency + two primes), 1-2 test houses, and a few component families. Mint tens of NFTs, not millions. The goal is to test governance roles, the payment flow, and whether labs actually publish.

3.2 Ideal adoption

In an ideal case, more labs publish reports by default, and procurement teams check the chain first before ordering tests. Over time, the network becomes a shared evidence layer. It does not need to replace every database, but it can become the standard place to anchor proofs.

3.3 Architecture pressures and expected changes

As usage grows, three pressures appear.

- State growth: many tokenIds and events. Solution: event indexing, light on-chain storage, and possibly L2 for high volume updates.
- Off-chain permanence: IPFS needs pinning. Solution: consortium pinning policy and optional Arweave backup for critical reports.
- Governance: more participants means more disputes and admission decisions. Solution: clear consortium rules; later, governance token is possible, but not needed for the pilot.